



PANDASIA
PREVENTING FUTURE PANDEMICS



Report on the parameterized model (M36)



A transdisciplinary research project with funded by the EU and the UKRI



Pandemic literacy and viral zoonotic spillover risk at the frontline of disease emergence in Southeast Asia to improve pandemic preparedness

PANDASIA

Report on the parameterized model

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Project Partners

No.	Abbreviations	Full names	Countries	Main roles
1	NMBU	Norwegian University of Life Sciences	Norway	Coordinator
2	NVI	Norwegian Veterinary Institute	Norway	WP4
3	UKHD	University of Heidelberg	Germany	WP5 lead
4	IZW	Leibniz Institute for Zoo and Wildlife Research	Germany	WP3-WP6
5	QMUL	Queen Mary University of London	UK	WP1
6	CU	Chulalongkorn University	Thailand	WP1 lead
7	UMU	Umeå University	Sweden	WP4 lead
8	KKU	Khon Kaen University	Thailand	WP3 lead
9	MU	Mahidol University	Thailand	WP2 lead
10	SUPA71	SUPA71 Co., Ltd.	Thailand	WP6 lead
11	NOVA	UNIVERSIDADE NOVA DE LISBOA	Portugal	WP9 lead



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2. The material presented and views expressed here are the responsibility of the author(s) only.
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List of Acronyms

EH	EcoHealth
EID	Emerging Infectious Disease
EU	European Union
GISAID	Global Initiative on Sharing All Influenza Data
GMS	Greater Mekong Subregion
HPH	Health Promoting Hospitals
ICF	Informed Consent Form
IDI	In-depth Interview
IPBES	Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services
LCLU	Land Cover - Land Use
MBDS	Mekong Basin Disease Surveillance
NIC	National Influenza Centres
IHR	International Health Regulations
OH	One Health
VHV	Village Health Volunteer
WHO	World Health Organization
WP	Work Package

Introduction

The COVID-19 pandemic has highlighted the importance of understanding local environmental, ecological, and societal drivers and barriers of zoonotic spillover risk. Spillover events arise at the human and animal interface – the frontline of zoonotic disease emergence – where humans interact with wildlife, livestock, insects, pathogens, etc., where pathogens have opportunities to jump to new species and initiate local emergence. Identifying the drivers of spillover and high-risk interfaces are critical for successful surveillance and developing interventions that can minimize spillover risk and improve overall human health outcomes.

There are at least 400 EIDs recognized as major public health issues, affecting millions of people globally, and 60% of these are zoonoses (1). The number of new emerging pathogens, including those originating in wildlife, has increased during the last 80 years even when accounting for improved surveillance (2, 3). Moreover, vector-borne diseases are a threat to 80% of the world's population and are estimated to be responsible for 17% of the global burden of communicable diseases and 30% of EID events (4).

The Greater Mekong Subregion (GMS) in Southeast Asia has been identified as a global hotspot with extraordinary biodiversity, including uncharacterized and unquantified viral diversity and a relatively high risk for emerging infectious diseases (EIDs), especially for zoonotic pathogens from wildlife and vector-borne pathogens and especially in areas with human encroachment into natural areas (2). The region is also a diversity hotspot for rhinophid bats, in which several novel SARS-CoV-2-related coronaviruses were found (5, 6). Further increasing the risk of emergence and catalyzing spread, a new high-speed rail line connecting China's Yunnan province with Thailand is operational in Laos and will soon extend to Bangkok as part of the Kunming–Singapore railway (Pan-Asia railway network) of China's Belt and Road Initiative. It is expected that personal travel will increase, subsequently increasing the risk for viral zoonotic spread. The region is, therefore, at risk for emerging infectious diseases and future pandemics that can have global implications and impacts on societies.

Understanding the risks is key to developing effective, proactive spillover prevention strategies. Our consortium consists of a highly qualified international team that integrates social, zoological, ecological, virological, public health, epidemiological, evolutionary, statistical and mathematical modelling perspectives with social-ecological systems theory and transdisciplinary participatory approaches. We apply our advances in prediction to determine, analyze and model viral pathogen spillover potential from OH and EH perspectives. The research and strategies will be based on community driven co-created interventions and communication strategies to improve pandemic health literacy and preparedness among local communities and public health authorities.

Project Objectives

The **main objective of PANDASIA** is to investigate potential pandemic drivers along nature-rural-urban gradients through a comprehensive collection of social and biological data and predictive modelling of zoonotic spillover rates and disease emergence in high-risk settings in Thailand. The project consists of seven work packages (WPs), as shown in Figure 1.

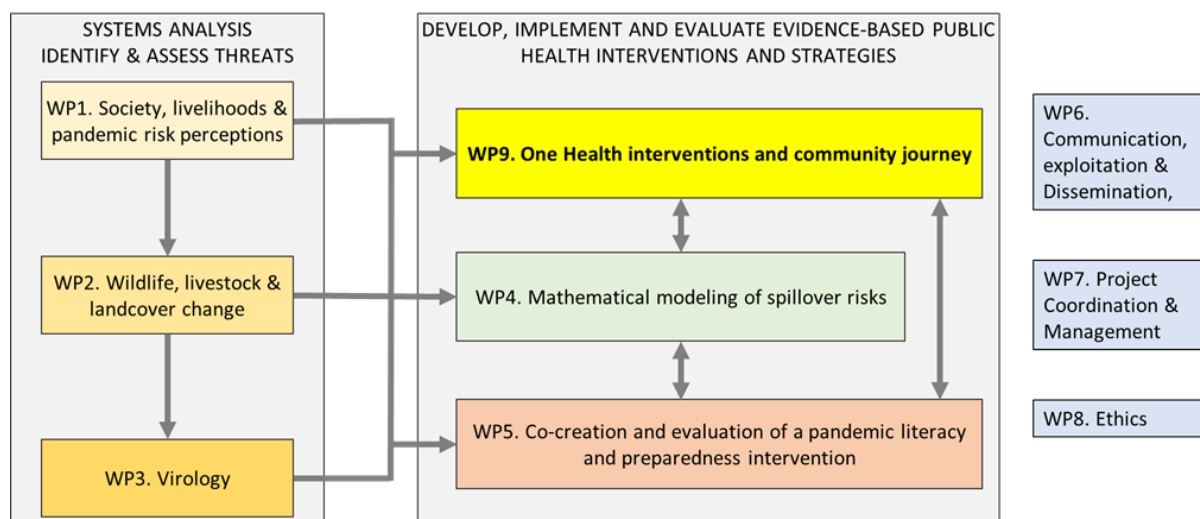


Figure 1. **PANDASIA Work Packages**

Objectives of Work Package 4

We strive to develop a new innovative model for spillover risk management building on local data from potential EID hotspot areas in Thailand. Empirical data on pathogens, environment, animal and social interconnectedness will be used to build a decision-making model and tools for users to learn from when innovating management strategies to inform surveillance and monitoring of relevance for pandemic preparedness. The specific objectives are to:

- 4.1** Develop a process-based model connecting natural environment, wild and domesticated animals, pathogens, and humans to spillover risk.
- 4.2** Parameterise the model using inverted neural networks and Bayesian inference and identification of drivers of spillover and pathogen evolution.
- 4.3** Develop user co-created scenarios for spillover risk management and pandemic preparedness monitoring and surveillance strategies across sectors.
- 4.4** Develop a decision-making tool for local pandemic preparedness

Overview of WP task status for deliverable

Task 4.1: Development of a model structure and theoretical analysis.

Lead: UMU, Co-lead: NVI.

We have developed multiple complementary process-based modelling frameworks that account for host population dynamics, host population mobility dynamics, disease transmission dynamics, within-host dynamics of pathogen strains, and evolutionary dynamics of pathogens; all unfolding across three landscape sectors: forest, rural and urban. These frameworks include both deterministic compartmental models and individual-based models, allowing analysis at multiple spatial and organisational scales.

This work has included systematic identification and formalisation of key ecological, epidemiological, and behavioural processes governing host populations and transmission pathways through integrative work between WP1–4. Evolutionary processes relevant to pathogen emergence and spillover, including host-shift and virulence dynamics, have been incorporated through joint work across WPs 1–4. By screening the salient parameter space, we are identifying regions in that space where spillover events become more likely, and which parameters and variables are more relevant to focus on.

Task 4.2 Data-driven modelling of ecological and evolutionary drivers of spillover

Lead: NVI, Colead: UMU.

We are parameterising relevant parts modelling framework developed in task 4.1 using a mix of machine learning, frequentist and Bayesian inference approaches. This involves combining data on process rates, environmental characteristics, theoretical values and field data to operate data-driven models within a realistic parameter space and form case studies applying our theoretical work on real-life cases. The models are providing estimates of spillover risk testable against incoming data and opening avenues for iterative improvements.

In parallel, foundational work has been carried out to enable data-driven model instantiation. This has included joint WP2–WP3–WP4 workshops and bilateral meetings to define parameterisation strategies for multi-species, multi-patch transmission models and evolutionary host-shift frameworks, aligned with available and forthcoming data streams. A preliminary integrated database has been assembled from WP1–3 datasets, and exploratory analyses of surveillance, capture, infection, and human survey data have been conducted to assess spatial and temporal patterns and identify candidate case-study systems. Building on this groundwork, a first data-driven stochastic differential equation (SDE) model has been implemented and parameterised for an initial case study using directly observed quantities (e.g. host counts, land-use structure, survey-derived contact rates) and conservative assumptions where data are incomplete. This implementation demonstrates the feasibility of translating conceptual models into data-informed stochastic representations and

provides a basis for structured scenario analysis and further refinement as additional data become available.

Deliverable 4.1: Report on the parameterized model (M36)

Approach

In our view disease emergence is a complex set of overlapping processes on different spatial and temporal scales. As PANDASIA focuses on viruses originating in wildlife and peri-domestic settings along an urbanization gradient and at important interfaces in the rural landscape, we focus on the interplay between ecological and evolutionary processes that drive disease emergence, structured around the following main elements:

- 1) Exposure links between the original host, bridge hosts, and humans
- 2) Host shift evolution as the potential pathogen adapts to new host(s)
- 3) Evolution of virulence as an emergent property of the host-pathogen environment

The main processes behind disease emergence are, to a large extent, considered as separate systems models on different scales, where some general modules can be developed separately and coupled together into a systems modelling framework for spillover risk assessment and intervention testing. As there is a practically infinite number of potential chains of events we aim for a balance between general underlying processes of the systems model approach and parameterising relevant case studies and enlightening examples.

We have also concluded that when modelling evolutionary processes, models need to:

- Be individual-focused on both the host and pathogen level
- Be time and space explicit to integrate processes on different scales
- Include environmental influence

Even though in some cases it is useful to calculate mean field dynamics.

For the emergence we have focused on a series of key processes (7-20):

- (1) Contact rates between wildlife and humans and how it is influenced by changes in land use and landscape fragmentation. An individual-based model of bat behaviour set in a landscape based on rural Thailand has investigated the effects of different landscape and resource structures on the exposure rate of humans to bat-borne pathogens.
- (2) Sensitivity of multi-host transmission pathways across human–domestic animal–wildlife interfaces. A multi-species, multi-environment transmission model has been analysed with sensitivity analyses to investigate how wildlife chronic stress, host movement between environments, and heterogeneous contact processes influence spillover risk. The framework represents transmission within and between wildlife, domestic animal, and human host populations across a small number of aggregated environment types and is designed to be extended to more spatially explicit settings.
- (3) Fundamentals of host shift evolution with regards to host compatibility, interspecific interaction, host density, and effects of host stress, translating as drivers on the risk of spillover process. Models representing host compatibility under different parameter sets have been investigated for evolutionary stability and direction of selection, identifying parameter spaces vulnerable to host shift evolution and thus connecting spillover to contact rate and host density.

- (4) Virulence as an emergent property of pathogen life history evolution, host properties and environmental impact. Notably, our current model focuses on viral strategies of “collaboration”, treating infections as non-clonal and finding the resulting possible stable states for an emerging infectious disease under given conditions, and how changes in host population structure and environment changes selection pressures for virulence, thus connecting spillover to disease causing potential signifying a spillover or emergence event as a pandemic threat.

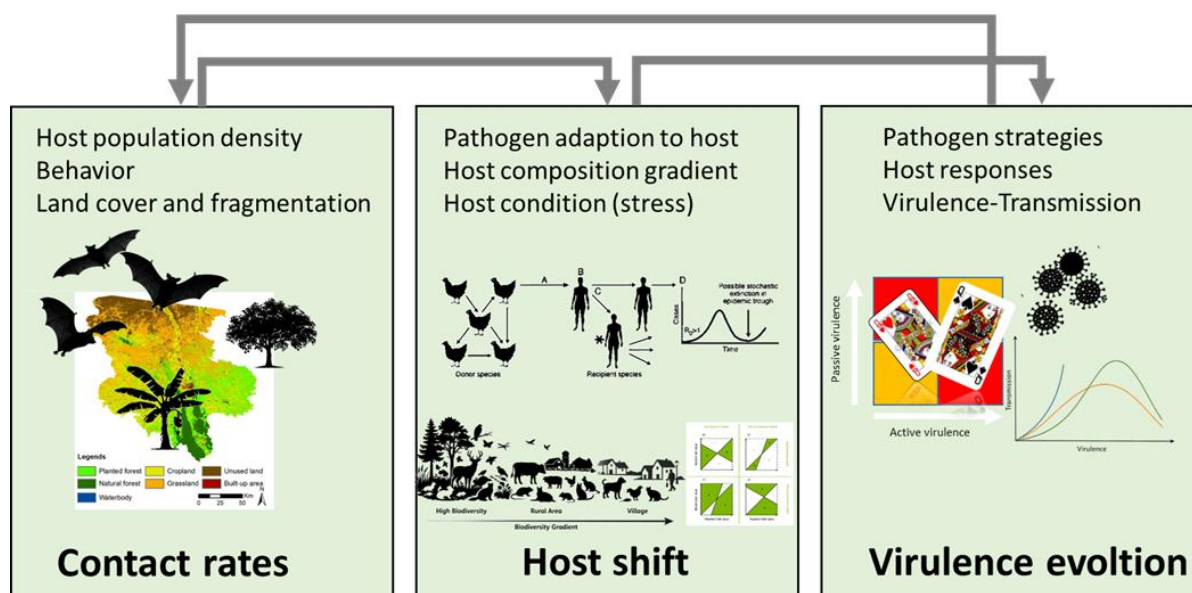


Figure 2. *Working model of disease emergence with parameterized component models*

Results summaries

The main results so far center on preliminary and finished output from the existing component models, which are being published in relevant scientific literature.

Contact rates: Two complementary modelling frameworks have been developed to estimate contact rates in settings where these are difficult to observe directly and to assess their influence on spillover risk at the system level.

(1) Individual-based bat movement model. Applying optimal foraging theory and individual-based modelling, we simulate flying-fox movement and population dynamics under a range of land-use and land-cover change (LCLUC) scenarios. The model has been parameterised using values from the literature and informed by empirical studies of bat behaviour and space use, and validated through qualitative comparison with observed movement and aggregation patterns. Using synthetic landscapes composed of urban, forest, orchard, and water-body habitats, we analyse the effects of agriculturalisation, urbanisation, forest fragmentation, forest degradation, and reforestation on bat densities and movement. The results indicate that habitat disruption—particularly fragmentation associated with urbanisation—leads to increased aggregation of bats near anthropogenic resources and elevates contact rates at boundaries between high human-density areas and habitats frequently used by bats. Forest degradation further exacerbates this effect, while reforestation reduces aggregation in human-dominated landscapes and lowers inferred spillover risk. Ongoing work focuses on extending the framework to new systems and applying Bayesian inference to estimate parameters that are not directly available from the

literature, thereby translating simulated movement outputs into quantitative contact-rate estimates suitable for downstream transmission models.

(2) Multi-species, multi-environment transmission framework. In parallel, a multi-species, multi-environment compartmental model has been developed and analysed to investigate how contact rates between wildlife, domestic animals, and humans, together with host movement between aggregated environments, influence spillover risk. This framework explicitly represents wildlife chronic stress as a modifier of susceptibility, transmissibility, and recovery, and was analysed using sensitivity analyses to identify key drivers of spillover dynamics.

Data-driven case-study with initial PANDASIA data: A first parameterised, data-driven case study was developed to demonstrate how the conceptual multi-host, multi-patch modelling framework can be translated into a stochastic formulation informed by empirical data generated within PANDASIA WPs 1–3. The case study focuses on zoonotic spillover risk in a heterogeneous rural landscape in Thailand and is formulated as a stochastic differential equation (SDE) system with susceptible–infected–susceptible (SIS) dynamics. The model represents transmission and host movement across three aggregated environment types corresponding to key interfaces in the study system, selected based on the availability of animal capture data, human visitation patterns, and reported contact behaviour from WPs 1–3.

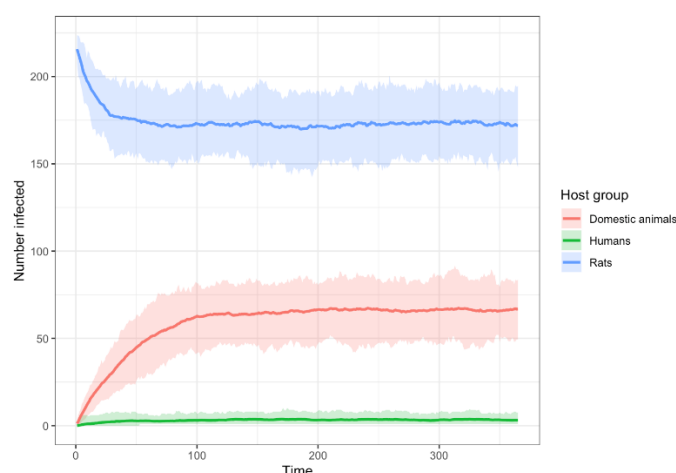


Figure 3. Stochastic runs of the simulation from parameterized model. Means and confidence intervals for 100 runs are shown. Infected populations summed for the same species across interfaces. Parameters for these simulations were derived from WP1-3 data where available.

The case study is focused on Poxviridae as the viral family of interest. This choice was motivated by its relatively high detection prevalence across multiple interfaces in the most frequently trapped host species (rats) in WP2 datasets, as well as by the biological plausibility of transmission within this viral family through close contact and shared environments. Although the model structure is general, transmission parameters and initial conditions in this implementation are interpreted specifically in the context of Poxviridae circulation.

Rodents (with emphasis on rats) constitute the primary wildlife host group in the model, reflecting the fact that the available virological and capture data originate from WP2 rodent

trapping. Domestic animals (cats and dogs) and humans were included as additional host groups based on two criteria derived from PANDASIA WPs 1–3: (i) their documented presence at the same interfaces as rats, as indicated by trapping records and survey data, and (ii) their reported direct and indirect contacts with humans, as captured through WP1 household and behavioural surveys. This structure enables representation of potential transmission pathways linking wildlife, peri-domestic animals, and humans at shared interfaces.

Initial host abundances, interface-specific distributions, and movement patterns were informed by WP2 capture data, WP1 demographic and survey data, and WP3 contextual information, with conservative assumptions applied where direct measurements were unavailable. Contact processes were structured to vary by host pair and environment type, with human–animal contact rates derived from WP1 survey responses and animal–animal contact rates represented using density-dependent formulations. The conceptual multi-environment model was converted into an SDE formulation to explicitly represent stochastic infection and recovery processes, enabling simulation of spillover dynamics under intrinsic variability.

Table 1. Example parameters in the parameterised multi-host, multi-patch SDE case study

Parameter	Description	Host(s) / Interface(s)	How parameter was defined or calculated	Data source (PANDASIA)
(N_r)	Initial rodent (rat) abundance	Rodents; all interfaces	Estimated from WP2 trapping counts, scaled to interface area; conservative extrapolation where sampling incomplete	WP2 (rodent trapping data)
(N_d)	Initial domestic animal abundance (cats, dogs)	Domestic animals; peri-domestic interfaces	Derived from household survey reports and village-level summaries; bounded by literature ranges	WP1 (household surveys), literature
(N_h)	Human population size	Humans; all interfaces	Taken from village demographic data and survey summaries	WP1 (demographic surveys)
($\beta_{r,h}$)	Rodent–human transmission coefficient (Poxviridae)	Rodent–human; all interfaces	Estimated from reported contact frequencies converted to per-capita daily rates; scaled by literature-based transmission plausibility	WP1 (contact surveys), literature
($\beta_{r,d}$)	Rodent–domestic animal transmission coefficient	Rodent–domestic animal; peri-domestic	Density-dependent formulation using rodent abundance and assumed contact kernels	WP2 (rodent abundance), literature
($\beta_{d,h}$)	Domestic animal–human transmission coefficient	Domestic animal–human; household	Estimated from reported pet–human contact frequencies	WP1 (contact surveys), literature
(γ_r)	Recovery rate (rodents)	Rodents	Set using pathogen–family–agnostic infectious period ranges relevant to Poxviridae	Literature
(γ_d)	Recovery rate (domestic animals)	Domestic animals	Literature-based ranges; explored via sensitivity analysis	Literature
(γ_h)	Recovery rate (humans)	Humans	Literature-based ranges; not inferred	Literature
(m_{ij}^{ij})	Human movement rate between interfaces (i \to j)	Humans; all interfaces	Estimated from WP1 survey data on visit frequency and time spent per interface	WP1 (movement & activity surveys)
($I_r(0)$)	Initial proportion infected (rodents)	Rodents	Calculated from Poxviridae detection prevalence in WP2 samples	WP2 (virological data)
($I_d(0)$)	Initial proportion infected (domestic animals)	Domestic animals	Set conservatively due to lack of direct data	Assumptions
($I_h(0)$)	Initial proportion infected (humans)	Humans	Set to zero (no assumed ongoing transmission)	Assumption

This parameterised SDE implementation constitutes a proof-of-concept data-driven model that meets the relevant deliverable requirements by demonstrating how empirical data from PANDASIA WPs 1–3 can be integrated into a stochastic spillover framework. While formal Bayesian inference or machine-learning–based parameter estimation has not yet been applied, the SDE formulation is explicitly designed to support these approaches in

subsequent work as additional datasets from PANDASIA and related collaborations become available.

Host shift:

(1) We have developed a multi-environment, multi-host compartmental model to identify key drivers of spillover risk across urban, rural, and natural environments. The framework incorporates heterogeneous host populations, inter-species contacts and host movement, and includes wildlife chronic stress as one factor influencing susceptibility, transmissibility and recovery. Sensitivity analyses show that domestic animal populations at agricultural interfaces seem to strongly amplify spillover risk through their recovery dynamics and per-contact transmission probabilities, while within-wildlife transmission across environments also shapes system dynamics. Chronic stress effects can further increase spillover risk, and the model indicates that reducing wildlife stress alongside improving domestic animal health and management can markedly lower the likelihood of cross-species transmission.

These findings highlight the importance of maintaining healthy wildlife populations and integrating physiological and ecological data into epidemiological modeling to inform both bottom-up interventions targeting domestic animals and top-down strategies that sustain resilient wildlife communities and thereby mitigate zoonotic spillover.

(2) In parallel, we developed a complementary multi-host, multi-pathogen strain stochastic model to investigate the interplay between host similarity, inter- and intraspecific interactions, and adaptive trade-offs in pathogen evolution, as well as the possibility of evolutionary host shifts—the evolutionary change in a pathogen’s natural host. The model framework allows pathogen strains to differ in their level of adaptiveness—quantified as the probability of infection success—across different host species. Specifically, pathogen strains are assumed to have a natural host for which they are specialized. Adaptiveness to other host species depends on the degree of difference between hosts. The natural host is assumed to be an inherent strategy, and the model allows mutant strategies to emerge through mutation in infected individuals. This enables us to track evolutionary changes in the pathogen population over time and to detect evolutionary host shifts.

We simulated pathogen evolution across a wide range of scenarios differing in host species similarity, population size, and intra- and interspecific contact intensity. Our results show that, while only a small proportion of potential evolutionary host-shift scenarios result in actual host shifts, higher levels of interspecific interaction increase host-shift opportunities, enabling host shifts between more dissimilar host species. These findings highlight the importance of interspecific interactions, such as wildlife exploitation and hunting, in the emergence of pathogens, and emphasize that these factors must be considered when designing intervention strategies.

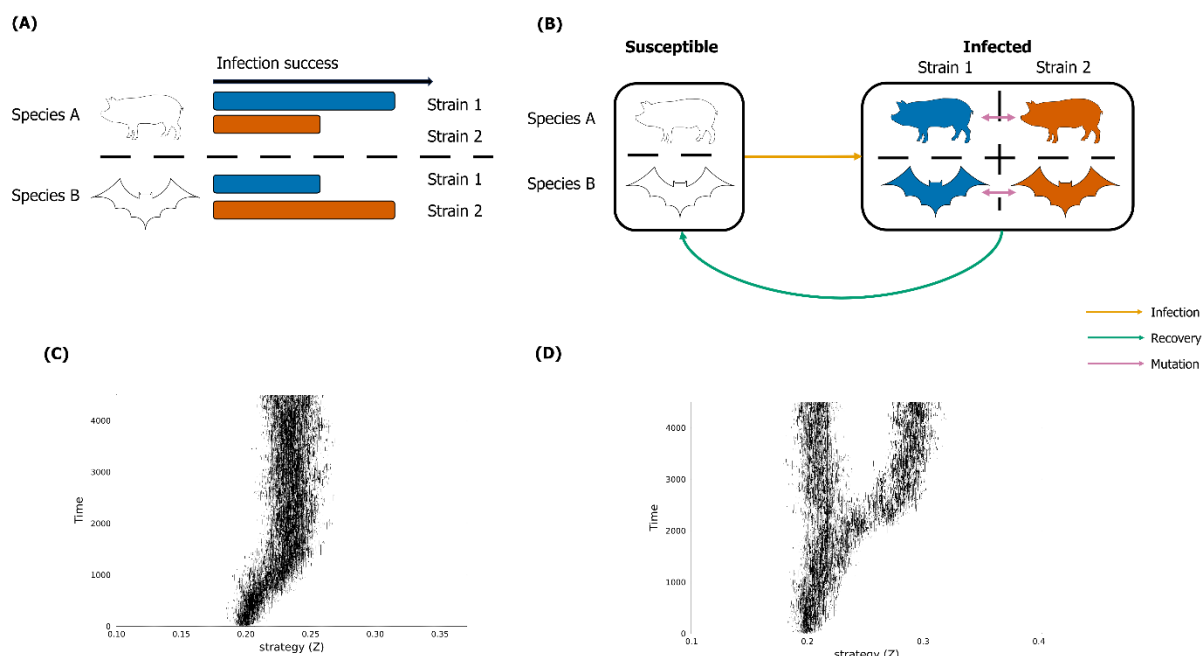


Figure 4. Conceptual schematic and simulations of the evolutionary host-shift framework. A system consisting of two host species (species A and B) and two pathogen variants (strains 1 and 2) is considered. **(A)** The two pathogen variants differ in their adaptivity to the two host species, quantified by infection success (graphically depicted by the length of the colored bars). **(B)** The host populations are subdivided into susceptible and infected individuals, and the infected population is further subdivided into individuals infected with strain 1 or strain 2. Susceptible individuals become infected through intra- or interspecific contact with infected individuals; the infection status is determined by the pathogen variant that causes the infection. Infected individuals return to the susceptible state after recovery. The pathogen causing the infection in an infected individual may change due to mutation. **(C)** and **(D)** depicts two simulations of the stochastic evolutionary host-shift model with two host species. Black indicates whether a specific pathogen variant (shown on the x-axis) exists in the system, i.e., whether at least one individual is infected with that variant, at a given time (shown on the y-axis). **(C)** Depicts a generalist outcome, in which the optimal strategy is adaptation to both host species. **(D)** Depicts an evolutionary branching outcome, in which the initially monomorphic pathogen population splits into two coexisting pathogen populations, each adapted to one of the host species.

Initial discussions have begun on extending the evolutionary host-shift framework to a data-driven model that utilizes the data collected and analyzed in this project. The work is in an early conceptual phase, and an initial proposed approach is to apply methods from studies of viral sharing (host–host) and virus–host association network data (see, for example, work by Albery et al. (21)). Specifically, a parametric data-driven model—such as generalized additive mixed-effects models (GAMMs)—would be used to model network data, created from data collected from project study sites, using explanatory variables such as phylogenetic distance between host species (21). The assumption is that the data-driven model, which can predict the probability of unobserved links in the network, can be used as a proxy for the probability of infection success for unobserved virus–host combinations. This would allow the incorporation of the data-driven model into the aforementioned evolutionary host-shift framework, enabling simulation of more site-specific host and environmental scenarios.

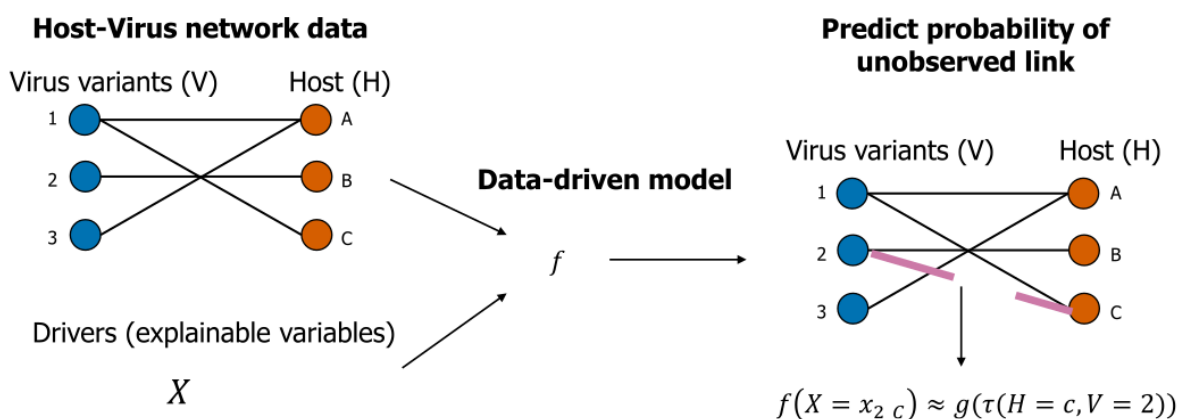


Figure 5. Conceptual schematic of a data-driven extension of the evolutionary host-shift framework. A parametric data-driven model, f , is trained to model the network structure and link probabilities of virus–host association network data or viral sharing (host–host) network data (not shown in the figure), using a set of explanatory variables X . The model f is assumed to be approximately monotonically related to the probability of infection success, $\tau(H, V)$, between a virus (V) and a host (H). The monotonic relationship between f and $\tau(H, V)$ is captured by a function g . In the figure, f is used to estimate the link probability for an unobserved association between host C and virus 2, denoted as $f(x_{2,C})$, where $x_{2,C}$ represents the explanatory variables for the C-2 pair. This is graphically depicted by the dashed line between C and 2 in the bipartite graph. The value $f(x_{2,C})$ is assumed to be approximately equal to $g(\tau(H = C, V = 2))$.

Virulence evolution: We developed models on the level of individual pathogen phenotypes embedded in variable host populations and environments, and investigated the emergent properties of traits selected for under different parameters for hosts, pathogens and environments. In general we see that most spillovers are not expected to necessarily be pathogenic, and may suffer a reduction in effective replication and emergent virulence and transmission from competition with the wider microbiome. Notably, the old paradigm of modelling virulence evolution by treating infections as clonal and optimizing on transmission or new infections caused in a naïve population can be very misleading and does not capture likely crucial mechanisms. Instead the newer approach of “sociovirology” where game theory on the level of genotypes and phenotypes and spatial structures on a smaller scale than the individual host is crucial to make sense of the otherwise seemingly random and bewildering pattern of disease-causing microorganisms evolving from a microbiome consisting of mostly weakly pathogenic, commensal or even mutualistic organisms.

The findings suggest that while complicated and situationally dependent, a new microorganism finding its way into the human-associated microbiome may respond predictably to changes in its environment, such as host conditions, immune responses, and to microbial diversity itself. This may help with EID risk analyses and mitigation strategies.

Impacts and Conclusion

The currently parameterized models are demonstrating a growing understanding of the pathways and mechanisms of disease emergence, but also the large number of possible pathways. Further work, both within PANDASIA and beyond this project, will be needed to further elucidate important pathways for different environments and different threats, balancing generalizability and specificity.

Progress in understanding emerging infectious viruses and their virulence in Southeast Asia can be systematically translated into improved pandemic preparedness both regionally and globally by improving targeted surveillance, research collaboration, and policy alignment across national governments and multilateral organizations.

At the regional level, Southeast Asian Ministries of Health may expand and integrate results into national plans for genomic surveillance to detect and characterize novel pathogens rapidly, drawing on partnerships with networks like the MBDS consortium, which facilitates cross-border data sharing, joint outbreak investigation, and workforce training among Cambodia, Laos, Myanmar, Thailand, Vietnam and China. Simultaneously NICs and IHR focal points sharing virus isolates and sequence data with global platforms such as GISAID to enable real-time tracking of mutations associated with increased virulence or transmission can give additional training, mitigation and preparedness information by being used on trained evolutionary models. By linking robust regional science and policy frameworks with global preparedness mechanisms, Southeast Asia can contribute to early detection, rapid response, and resilient health systems capable of mitigating the next pandemic threat.

Further progress

The system of parameterized models will be expanded, validated, and further adapted for use with the data from this project as well as external data. Formal publications of the first parameterized models have been submitted and will be part of the PANDASIA catalogue once the scientific publication process is complete for each component.

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